

ACRME Hard Constraints Reference

Classification: Principal Cloud Architect — Placement Engine Design

Version: 1.0

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Status: Production Ready

v2.4 reconciliation note (7 Sep 2026). Requirements Baseline **v2.4** adds a new hard constraint **HC-11 AVAILABILITY_SET_INELIGIBLE (CAP-020)**: Availability-Set VMs are ineligible for zonal on-demand capacity reservations and are rejected from the zonal reservation / per-AZ CRG path at onboarding. It pairs with the CAP-021 deallocate/redeploy-to-AZ onboarding precondition. See HC-11 in Part 1 and the summary table in Part 2; normative detail is in ADR-002 v2.4 and TDD §8.1. All prior hard constraints (HC-1..HC-10) are unchanged.

v2.2 reconciliation note (2 Sep 2026). Under Requirements Baseline v2.2 and ADR-002 v2.2 the **single governed quota pool** is the **primary** model: one pool per region/quota family covers Prod + NonProd/CVAL + DR, with Prod and DR protected by **logical earmarks** (`Prod_Reserved_Floor` , `DR_Earmark_vCPU`) rather than physical group separation. The **Two-Group Quota Architecture** referenced by HC-3 and HC-7 below is retained as the sanctioned **exception topology** (used only when Azure Quota Group limits or a mandatory Prod-isolation boundary make one pool impossible). The HC-3/HC-7 arithmetic is unchanged and remains correct; in the single-pool model the terms map as `Effective_NonProd_Ceiling` → `Allocatable_NonProd` , `NonProd_DR_Group_Limit` → `Pool_Limit` , and `DR_Floor_vCPU` → `DR_Earmark_vCPU` (max-not-sum, DR-017). Also: the EU cross-geo DR extension region is **Switzerland North** (REG-002). See ADR-002 v2.2, the Calculation Logic Reference v2.2 (Scenario 8/9), the FDD Section 4.2, and the TDD Section 8.3.

✓ **[Amended v2.4] Middle East DR — now offered cross-geo into Europe (DEC-001 RESOLVED).** Per the latest legal update, **DR is now provided for the Middle East**. Prod and CVAL are provisioned in a weighted-selected **Middle East** Standard region (separate CRGs per ENV-003), and **DR is placed cross-geo in a weighted-selected Europe Standard region** — Europe being an approved DR destination for the Middle East (assumption **A-ME1**; per-country data-residency carve-outs are config-driven and out of engine scope). Both selections use the **weighted capacity model** (DR-020, PLC-010b, REG-002/REG-003). This supersedes the former `DR_NOT_OFFERED` position (the generic DR-014 flag remains available for any future geography). Consequences for placement below:

- **A Middle East seed now records a real cross-geo `dr_region` in a weighted-selected Europe region.** The engine creates a `SourceDestinationDRIndex` entry and reserves a DR earmark at the Europe destination.
- **The Europe DR region is weighted-selected among Europe Standard regions** (Switzerland North is the default example, not a fixed region). Changing the Europe destination set is a config change, no code change (REG-001/REG-002, HC-10).
- **The Europe destination sizes DR max-not-sum (DR-017)** across all sources failing over to it, including Middle East sources — this adds DR load to Europe.
- **CVAL co-locates with Prod locally (not with DR)**, so the CVAL-sacrifice DR bootstrap (DR-005/006) does **not** apply to the Middle East; ME DR uses dedicated reserved capacity in Europe. See ADR-001, ADR-005, the FDD Section 4.4/Section 8, the TDD Section 2.2/Section 9, and the Calculation Logic Reference Scenario 4.

Overview

This document consolidates all **Hard Constraints (HC-1 through HC-11)** governing regional placement and reservation-eligibility decisions in the **Azure Capacity Reservation Management Engine (ACRME)**. Hard constraints are **binary pass/fail gates** — any region failing a region-level hard constraint is excluded from scoring entirely, not penalized with a lower score; the VM-level eligibility gate **HC-11** (v2.4) excludes ineligible VMs from the zonal reservation path at onboarding.

Enforcement Architecture

Hard constraints are enforced at two stages in the placement decision pipeline:

Stage 1 — Pre-filtering (before candidate set formation):

- **HC-8** GEOGRAPHY_CONTAINMENT
- **HC-9** STANDARD_REGION_ONLY

Stage 2 — Hard Constraint Gate (after geography/class filtering, before scoring):

- **HC-1** REGION_SEPARATION
- **HC-2** CAPACITY_FLOOR
- **HC-3** QUOTA_FLOOR
- **HC-4** DR_SEPARATION_CLASS
- **HC-5** ZONE_AVAILABILITY
- **HC-6** DR_COVERAGE_FLOOR
- **HC-7** DR_FLOOR_INTEGRITY
- **HC-10** CROSS_GEO_EXTENSION_PATH_APPROVED

Regions surviving both stages enter the **scoring pipeline** where soft objectives (`PS_Prod` , `PS_NonProd` , `PS_DR`) rank them.

Source Documents

- **HC-1 through HC-7:** `multi_region_placement_design.md` Section 27 (Hard Constraints)
- **HC-8, HC-9, HC-10:** `acrme_production_readiness_review_and_architecture.md` Section 27 (Production-Ready Region Selection)
- **Design rationale:** `design_change_summary.md` (Decisions D1-D9)

Part 1 — Hard Constraint Definitions

HC-1: REGION_SEPARATION [UPDATED]

Category: Isolation

Source: Multi-Region Placement Design Section 27; Decision D8

Status: Updated in Pass 2 — NonProd/DR co-location now permitted

Definition

```
NonProd_region ≠ Prod_region
DR_region      ≠ Prod_region
[Constraint DR_region ≠ NonProd_region REMOVED – NonProd and DR may share a region]
```

Rationale

NonProd/DR co-location is permitted to allow **DR overflow capacity reuse** from the NonProd CRG. This change (Decision D8) enables the engine to leverage unused NonProd capacity as a DR buffer, reducing the minimum region count from 4 to 3 — and, in the two-region model, to 2.

The number of regions available in a geography (a configurable value, REG-001) determines how the environments distribute:

With 2 regions (co-located model for Europe, Australia, Asia Pacific):

Prod is isolated in one region; **NonProd/CVAL and DR co-locate in the other region — deterministically and mandatorily** (there is no third region to separate them). This is the normal outcome, not an exception (baseline PLC-010a). VR-1 (Prod $\neq$ DR) and VR-2 (Prod $\neq$ CVAL) still hold because Prod is in the other region; VR-3 (CVAL/DR co-location) is satisfied by design. HC-6/HC-7 combined-capacity checks apply to the shared co-located region. **[Amended v2.4]** (The Middle East does not follow this co-located pattern: Prod and CVAL co-locate in a Middle East region and DR is placed cross-geo in a weighted-selected Europe region — PLC-010b, DR-020, DEC-001 RESOLVED.)

With 3 regions (current model for US):

Prod is isolated; NonProd and DR both draw from the remaining 2 regions (they may land on the same region or on different ones — determined by HC-6 and PS score).

With 4 regions:

Prod eliminates 1; NonProd selects from 3; DR may share with NonProd or use the remaining regions — whichever satisfies HC-6 and maximises PS_DR.

Implementation

- Prod region is excluded from NonProd candidate set
- Prod region is excluded from DR candidate set
- NonProd region is **not** excluded from DR candidate set (co-location allowed)
- CVAL (alias for NonProd) and Prod must not share a region
- CVAL and DR may share a region only under the approved policy (HC-1 update per D8)

Validation Rules (VR)

- **VR-1:** Prod and DR must not share a region
- **VR-2:** CVAL and Prod must not share a region
- **VR-3:** CVAL and DR may share a region only under the approved policy

Evidence Tag

[Derived] from Azure Well-Architected Framework (Reliability pillar)

HC-2: CAPACITY_FLOOR

Category: Capacity

Source: Multi-Region Placement Design Section 27

Status: Stable

Definition

CR headroom in candidate region $\geq$ minimum_reservation_units

where:

minimum_reservation_units = $2 \times \text{requested_vm_count} \times \text{SKU_vCPU_count}$ (default multiplier: 2)

CR headroom = total_free_slots (in the target CRG type: Prod/NonProd/DR)

Rationale

Ensures sufficient capacity buffer exists in the region before placement. The **2x multiplier** provides headroom for growth and prevents placement into near-exhaustion regions.

Implementation

- Read `RegionalSnapshot.prod_crg_free_slots`, `nonprod_crg_free_slots`, or `dr_crg_free_slots` (per environment type)
- Compare against `minimum_reservation_units` (configurable per PlacementPolicy)
- **Regions at capacity floor are excluded from scoring entirely**

Configuration

- **Default multiplier:** 2
- **Configurable via:** `PlacementPolicy.capacity_floor_multiplier`
- **Overridable per geography or SKU class**

Evidence Tag

[Undocumented – architectural judgement]

HC-3: QUOTA_FLOOR [UPDATED]

Category: Quota

Source: Multi-Region Placement Design Section 27; Design Change Summary QG-3

Status: Updated in Pass 2 — now reads from Quota Groups, not per-subscription QuotaRecord

Definition

For Prod placement in region R:
 REJECT if: $\text{Prod_Group_headroom}(R) < \text{requested_vm_count} \times \text{vCPU_per_instance}$

For NonProd placement in region R:
 REJECT if: $\text{nonprod_headroom}(R) < \text{requested_vm_count} \times \text{vCPU_per_instance}$
 (nonprod_headroom uses effective_nonprod_ceiling – floor-adjusted, not raw group_limit)

For DR placement in region R:
 REJECT if: $\text{dr_headroom}(R) < \text{target_dr_qty} \times \text{vCPU_per_instance}$
 (target_dr_qty = customer's configured DR bootstrap target – DR-007/ENV-005,
 configurable per workload/product/region; NOT a fixed dr_ratio. Destination-region
 former fixed DR floor is sized max-not-sum: MAX(source portions), A.6/DR-017. The
 30–40% dr_ratio model is superseded – see Calc Logic Reference Scenario 15 → 17.)

Data Source

- **Primary:** QuotaGroup entity (Cosmos DB), synced to RegionalSnapshot (Redis)
- **Fields:**
 - $\text{Prod_Group_headroom}(R)$ = Prod group available quota in region R
 - $\text{nonprod_headroom}(R)$ = NonProd+DR group available quota **minus DR floor** (see HC-7)
 - $\text{dr_headroom}(R)$ = DR portion of NonProd+DR group quota
 - **Legacy field:** QuotaRecord.deployment_headroom is superseded for placement checks; retained for per-subscription quota monitoring only

Formulas

Prod Group Headroom:

$$\text{Prod_Group_headroom}(R) = \text{Prod_Group_Limit}(R) - \text{Prod_Group_Used}(R)$$

where:

$$\text{Prod_Group_Limit}(R) = \text{base_subscription_quota_limit} \times (1 + \text{emergency_transfer_headroom_vcpu} / \text{base_limit})$$

$$\text{Prod_Group_Used}(R) = \sum (\text{prod_crg.quantity} \times \text{vCPU}) \text{ for all Prod CRGs in region R}$$

NonProd Headroom (effective ceiling):

$$\text{nonprod_headroom}(R) = \text{effective_nonprod_ceiling}(R) - \text{nonprod_quota_used}(R)$$

where:

$$\text{effective_nonprod_ceiling}(R) = \text{NonProd_DR_Group_Limit}(R) - \text{DR_Floor_vCPU}(R)$$

$$\text{NonProd_DR_Group_Limit}(R) = \text{base_subscription_quota_limit} \times (1 + \text{emergency_transfer_headroom_vcpu} / \text{base_limit})$$

$$\text{DR_Floor_vCPU}(R) = \text{Destination_DR_Requirement}(R) \times \text{vCPU} \quad \# \text{ max-not-sum (A.6/DR-017); NOT potential_dr_demand} \times \text{dr_ratio_max}$$

$$\text{nonprod_quota_used}(R) = \sum (\text{nonprod_crg.quantity} \times \text{vCPU}) \text{ for all NonProd CRGs in region R}$$

DR Headroom:

$$\text{dr_headroom}(R) = \text{DR_Floor_vCPU}(R) - \text{dr_quota_used}(R)$$

where:

$$\text{dr_quota_used}(R) = \sum (\text{dr_crg.quantity} \times \text{vCPU}) \text{ for all DR CRGs in region } R$$

Validation Sequence

1. **HC-3 (Quota Floor):** Check raw headroom availability
2. **HC-7 (DR Floor Integrity):** Check floor compliance (applies only to NonProd placements)

POC Blocker

B-1: Azure Quota Groups functionality must be GA-available in target tenant/region. If `Microsoft.Capacity/resourceProviders/.../groupQuotas` returns 404, escalate to Azure Support.

POC-30 gates this.

Evidence Tag

[Documented] — Azure Quota Groups Preview API documentation

HC-4: DR_SEPARATION_CLASS

Category: Resilience

Source: Multi-Region Placement Design Section 27

Status: Stable (non-paired regions only)

Definition

DR_region must have Separation_Class(DR_region, Prod_region) = HIGH

Rationale

Ensures non-correlated failure domains between Prod and DR regions. This constraint applies **only to non-paired regions** (e.g., standalone configuration with 3-4 explicitly chosen regions). Paired regions (e.g., East US / West US) are assumed to have HIGH separation by Azure design.

Separation Class Matrix

Prod Region	DR Region	Separation Class	Eligible?
East US	West US	HIGH	✓ Yes (paired)
East US	Central US	MEDIUM	✗ No
East US	West Europe	HIGH	✓ Yes (cross-geo)
UAE North	Saudi Arabia Central	MEDIUM	✗ No (same geography)
UAE North	Switzerland North	HIGH	✓ [Amended v2.4] Active cross-geo DR path — Middle East DR into a weighted-selected Europe region (Switzerland North is the default example); DR-020, PLC-010b, DEC-001 RESOLVED

Implementation

- Read `SeparationClassRegistry` (static config or Azure metadata service)
- For non-paired regions, validate `Separation_Class(DR, Prod) = HIGH`
- Paired regions bypass this check (assumed HIGH by default)

Evidence Tag

[Undocumented – architectural judgement]

HC-5: ZONE_AVAILABILITY

Category: Resilience

Source: Multi-Region Placement Design Section 27

Status: Stable

Definition

Candidate region must have ≥ 2 physical Availability Zones

Rationale

Single-zone regions are excluded for all non-dev environments. Multi-zone deployment is a **Well-Architected Framework Reliability pillar** requirement.

Implementation

- Read `RegionalSnapshot.zone_count` (populated from Azure metadata)

- Reject regions where `zone_count < 2`
- **Applies to:** Prod, CVAL, DR placements
- **Exemption:** Dev environments may bypass this constraint (configurable via `Placement-Policy.allow_single_zone_dev`)

Examples

Region	Zone Count	Eligible?
East US	3	✓ Yes
West Europe	3	✓ Yes
Central India	3	✓ Yes
Brazil South	1	✗ No
West US 3	3	✓ Yes

Evidence Tag

[Undocumented – architectural judgement]

HC-6: DR_COVERAGE_FLOOR [NEW]

Category: DR Capacity

Source: Multi-Region Placement Design Section 27; Design Change Summary CR/CRG-1; Decision D8

Status: New in Pass 2 — operationalises NonProd/DR co-location

Definition

REJECT DR placement in region R if combined DR + NonProd overflow capacity is insufficient to absorb the new customer's DR demand:

$$\text{dr_crg_free_slots}(R) + \text{nonprod_crg_effective_free}(R) < \text{customer_requested_dr_slots}$$

where:

`customer_requested_dr_slots` = customer's configured DR bootstrap target (`dr_bootstrap_qty`)

`nonprod_crg_effective_free` = `nonprod_crg_free_slots` - `nonprod_crg_dr_overflow_reserve`

`dr_bootstrap_qty` = configurable DR bootstrap target per workload/product/region (DR-007, ENV-005, C-1) – "enough to bootstrap; used-is-free". There is NO fixed DR-to-Prod ratio. The destination-region DR floor is sized max-not-sum (`Destination_DR_Requirement(R) = MAX(source portions), A.6/DR-017`). The former fixed `dr_ratio` (0.30–0.40) model is superseded – Calc Logic Reference Scenario 15 → Scenario 17.

Rationale

Since DR and NonProd may share a region (HC-1 constraint removed per D8), the engine must verify that the **combined capacity pool** can absorb the customer's DR need — either from the DR CRG directly, or from the NonProd CRG's reserved overflow headroom.

This constraint operationalises HC-1 co-location at the capacity selection level.

Data Requirements

Per-CRG-type snapshot fields (added in Pass 2):

- `nonprod_crg_free_slots(R)` — available NonProd CR capacity
- `nonprod_crg_effective_free(R)` — NonProd capacity minus DR overflow reserve
- `nonprod_crg_dr_overflow_reserve(R)` — capacity earmarked for DR failover absorption
- `dr_crg_free_slots(R)` — available DR CR capacity

Implementation Pseudocode

```
def validate_HC6_DR_coverage_floor(region, customer_dr_bootstrap_qty):
    snapshot = get_regional_snapshot(region)

    # DR demand is the customer's configured bootstrap target (DR-007/ENV-005),
    # NOT prod_vm_count × a fixed dr_ratio. The destination DR floor is sized
    # max-not-sum: MAX(source portions) per A.6/DR-017 (see Scenario 17).
    customer_requested_dr_slots = customer_dr_bootstrap_qty

    combined_dr_capacity = (snapshot.dr_crg_free_slots +
                           snapshot.nonprod_crg_effective_free)

    if combined_dr_capacity < customer_requested_dr_slots:
        return REJECT, "HC-6 DR_COVERAGE_FLOOR: Insufficient combined DR+NonProd
overflow capacity"

    return PASS
```

Policy Constants

- **dr_bootstrap_qty:** configurable DR bootstrap target per workload/product/region (DR-007, ENV-005, C-1) — “enough to bootstrap; used-is-free”. **No fixed DR-to-Prod ratio.**
- **Destination DR sizing:** max-not-sum — `Destination_DR_Requirement(R) = MAX(source portions failing over to R)` (A.6 / DR-017 / Calc Logic Reference Scenario 17). A per-scope `SUM` override is available only where a contract requires concurrent-failure protection (C-11).

Superseded — do not use. The former fixed `dr_ratio_min/max/target` (0.30/0.40/0.35) DR-to-Prod ratio model is **retired** (Calc Logic Reference Scenario 15). It produced idle reserves estimated at **\$1.5M-\$5M/year** at platform scale (Requirements Baseline Appendix D) and is no longer used in any HC-6/HC-7 calculation.

Evidence Tag

[Derived] from Decision D8 (NonProd/DR co-location); DR sizing per max-not-sum (Scenario 17, DR-017)

HC-7: DR_FLOOR_INTEGRITY [NEW]

Category: Quota Protection

Source: Multi-Region Placement Design Section 27; Design Change Summary QG-4

Status: New in Pass 2 — enforces DR floor within Two-Group Quota Architecture

Definition

REJECT NonProd placement in region R if placing the requested quantity would push `nonprod_quota_used` above `effective_nonprod_ceiling`:

$$\text{nonprod_quota_used}(R) + (\text{requested_vm_count} \times \text{vCPU_per_instance}) > \text{effective_nonprod_ceiling}(R)$$

where:

$$\text{effective_nonprod_ceiling}(R) = \text{NonProd_DR_Group_Limit}(R) - \text{DR_Floor_vCPU}(R)$$

Rationale

Prevents NonProd allocation from **encroaching on the protected DR quota floor**. Within the Two-Group Quota Architecture, the NonProd+DR group has a shared limit; the DR floor reserves a portion for DR use only. (Under the v2.2+ single governed quota pool the same arithmetic applies with `NonProd_DR_Group_Limit` → `Pool_Limit` and `DR_Floor_vCPU` → `DR_Earmark_vCPU`.)

Two-region geographies. In any geography with only two Standard regions (currently Europe, Australia, Asia Pacific, Middle East), CVAL/NonProd and DR are co-located in the single non-Prod region (PLC-010a). HC-7 is therefore especially load-bearing there: it — together with HC-6 — is what guarantees the shared co-located pool can carry the customer's DR demand without the NonProd allocation eroding the DR earmark. This check is geography-agnostic and evaluated against whichever region the two-region placement co-locates onto.

Evaluation Order

1. **HC-3 (Quota Floor):** Check raw headroom (`nonprod_headroom` ≥ `requested`)
2. **HC-7 (DR Floor Integrity):** Check floor compliance (`used` + `requested` ≤ `effective_ceiling`)

HC-3 checks availability; HC-7 checks compliance.

Formulas

Effective NonProd Ceiling:

$$\text{effective_nonprod_ceiling}(R) = \text{NonProd_DR_Group_Limit}(R) - \text{DR_Floor_vCPU}(R)$$

DR Floor (per region) — max-not-sum (A.6 / DR-017):

$$\text{DR_Floor_vCPU}(R) = \text{Destination_DR_Requirement}(R) \times \text{vCPU_per_instance}$$

where:

$$\text{Destination_DR_Requirement}(R) = \text{MAX}(\text{source_portion}(s) \text{ for every Prod CRG that could fail over to } R)$$

non-concurrent single-failure assumption (DR-001):
 # size to the LARGEST single source, not the sum.
 # each `source_portion` is that source's configured DR

bootstrap

quantity (DR-007) — `not_prod_qty` × a fixed ratio.

Superseded — do not use. The former $\text{DR_Floor_vCPU}(R) = \text{potential_dr_demand}(R) \times \text{vCPU} \times \text{dr_ratio_max}$ (SUM of sources $\times$ fixed 0.40) is **retired**. The destination DR floor is now the largest single protected source (max-not-sum), not the sum of all sources scaled by a fixed ratio (Calc Logic Reference Scenario 15 $\rightarrow$ Scenario 17). A per-scope **SUM** override (C-11) remains only for contractual concurrent-failure protection.

POC Example (GP-06 Topology)

Inputs:

- Prod demand: 80 VMs $\times$ 2 vCPU = 160 vCPU
- Base subscription quota limit: 200 vCPU
- Sources that could fail over to this region R, with their configured DR bootstrap portions (DR-007):
- Source A (this customer): **32 VMs** $\leftarrow$ largest source
- Source B: 24 VMs

Group Limits:

- **Prod Group:** $200 \times 1.28 = 256$ vCPU (28% emergency headroom)
- **NonProd+DR Group:** $200 \times 1.28 = 256$ vCPU

DR Floor Calculation (max-not-sum, A.6/DR-017):

```
Destination_DR_Requirement(R) = MAX(32 VMs, 24 VMs) = 32 VMs    # NOT SUM = 56 VMs
DR_Floor_vCPU = Destination_DR_Requirement(R)  $\times$  vCPU_per_instance = 32 VMs  $\times$  2 vCPU = 64 vCPU
# (Under the retired fixed-ratio model this would have been 80 VMs  $\times$  2 vCPU  $\times$  0.40 = 64 vCPU
# for a single source, but SUM-scaling multiple sources over-provisions – see Scenario 17.)
```

Effective NonProd Ceiling:

```
effective_nonprod_ceiling = 256 - 64 = 192 vCPU
```

Enforcement:

```
If nonprod_quota_used = 150 vCPU:
  New NonProd request for 60 vCPU  $\times$  2 = 120 vCPU
  150 + 120 = 270 > 192 ✗ REJECT (HC-7 violation)

If nonprod_quota_used = 100 vCPU:
  New NonProd request for 40 vCPU  $\times$  2 = 80 vCPU
  100 + 80 = 180  $\leq$  192 ✓ PASS
```

POC Blocker

B-2: Quota pool release behavior on CR reduction (Tier 2/3 quota-neutral claim) must be validated.

POC-31 measures release latency; confirm < 5 min for Tier RTO compliance.

Evidence Tag

[Derived] from Two-Group Quota Architecture design

HC-8: GEOGRAPHY_CONTAINMENT [IMPLIED]

Category: Geography

Source: Production Readiness Review Section 27

Status: Implied constraint (no standalone definition block)

Definition

In Scenario 1 (geography-based placement), derived Prod region must be within the Standard Capacity Regions for the customer's chosen geography.

Rationale

When a customer supplies an Azure geography (one of the five in-scope geographies — **US, Europe, Australia, Asia Pacific, Middle East**) rather than a specific region, the engine derives the Prod anchor via `argmax(PS_Prod)` over Standard Capacity Regions **within that geography only**. This constraint ensures the derived Prod region respects the customer's geographic preference.

The set of valid geographies and their Standard/Restricted region membership is **configuration-driven (REG-001)** — the engine must read the in-scope catalogue from `PlacementPolicy` rather than hard-coding a fixed geography list, so that adding/removing a geography or region is a config change, not a code change.

Enforcement Stage

Stage 1 — Pre-filtering (before candidate set formation)

Validation Rules (VR)

- **VR-4:** Derived Prod region must be within the Standard Capacity Regions for the customer's chosen geography

Implementation

```
def get_prod_candidates(geography):
    all_regions = get_all_azure_regions()

    # Stage 1: Geography containment (HC-8)
    geo_filtered = [r for r in all_regions
                    if r.geography == geography and
                       r.region_class == "Standard"]

    # Stage 2: Apply HC-1..HC-7, HC-10
    hc_filtered = apply_hard_constraints(geo_filtered)

    # Stage 3: Score and rank
    return argmax(hc_filtered, key=lambda r: PS_Prod(r))
```

Examples

Customer Geo- graphy	Distribution model	Eligible Standard Regions	Ineligible (HC-8)
US	3-region	West US 3, Central US, Canada Central	East US 2 (Restricted), all non-US regions
Europe	2-region	Switzerland North, Sweden Central	North Europe, West Europe (Restricted), all non-EU regions
Australia	2-region	Australia East, Australia Southeast	all non-AU regions
Asia Pacific	2-region	East Asia, Southeast Asia	Japan East (pending confirmation), all non-APAC regions
Middle East	[Amended v2.4] cross-geo DR	UAE North, Saudi Arabia Central (Prod+CVAL) + weighted-selected Europe region (DR)	East US*

Europe Standard regions become eligible for the Middle East as DR destinations only (not for Prod) via the HC-10 Cross-Geo Extension Path*, weighted-selected (DR-020, PLC-010b, DEC-001 RESOLVED).

Region lists above are examples of the current configured catalogue (Requirements Baseline v2.3 Section 6). The authoritative, versioned lists live in `PlacementPolicy (REG-001)`; the engine must not hard-code them.

Evidence Tag

[Undocumented – architectural judgement]

HC-9: STANDARD_REGION_ONLY [NEW]

Category: Region Class

Source: Production Readiness Review Section 27

Status: New gate for production-ready region classification

Definition

Automated placement, scoring, recommendation, and all environment assignments (Prod, CVAL, DR) must use Standard Capacity Regions only.

Rationale

Restricted Capacity Regions (sovereign clouds, government regions, regions with SKU limitations) are excluded from automated placement. Their exclusion is enforced at **Stage 1 of the eligibility decision tree**, not as a scoring penalty.

Exception deployments (customer explicitly requests a Restricted region) proceed via the **Scenario 2 exception path** only, requiring operator approval.

Region Classification

Standard Capacity Regions:

- Commercial Azure public cloud regions with full SKU availability
- No sovereign cloud restrictions
- No special approval required for CRG creation
- Eligible for automated placement of Prod, CVAL/NonProd, and DR
- In-scope catalogue examples (Baseline v2.3 Section 6): West US 3, Central US, Canada Central (US); Switzerland North, Sweden Central (Europe); Australia East, Australia Southeast (Australia); East Asia, Southeast Asia (Asia Pacific); UAE North, Saudi Arabia Central (Middle East)

Restricted Capacity Regions:

- Azure Government (US DoD, US Gov)
- Azure China (21Vianet)
- Regions with limited SKU availability (flagged in Azure metadata)
- Regions requiring special tenant approval (sovereign cloud onboarding)
- In-scope catalogue examples flagged Restricted: East US 2 (US); North Europe, West Europe (Europe)
- Prod-only via the Scenario 2 exception path; **CVAL and DR must never use them (VR-6)**

Pending / conditional: Japan East is under business review for inclusion as an Asia Pacific Standard region; until confirmed in `PlacementPolicy` it is not selectable by automated placement.

Enforcement Stage

Stage 1 — Pre-filtering (before candidate set formation)

Validation Rules (VR)

- **VR-5:** All automated placement paths (Scenario 1 geography-based, Prod derivation, NonProd/DR selection) must use Standard Capacity Regions only
- **VR-6:** CVAL and DR must not use Restricted Capacity Regions under any condition, including exception deployments

Implementation

```
def get_eligible_regions_stage1(geography, customer_request):
    all_regions = get_all_azure_regions()

    # HC-9: Standard regions only (Stage 1 pre-filter)
    standard_regions = [r for r in all_regions if r.region_class == "Standard"]

    # If customer explicitly requests a Restricted region → Scenario 2 exception path
    if customer_request.region_override and customer_request.region_override.region_class == "Restricted":
        return handle_exception_deployment(customer_request)

    # HC-8: Geography containment
    geo_filtered = [r for r in standard_regions if r.geography == geography]

    return geo_filtered
```

Exception Deployment Workflow (Scenario 2)

Trigger: Customer explicitly requests a Restricted Capacity Region

Workflow:

1. Operator receives request via exception deployment queue
2. Validate customer has required tenant permissions (sovereign cloud registration)
3. Validate SKU availability in target Restricted region
4. Manual approval gate (ops team review)
5. If approved: Create Prod CRG in Restricted region; **CVAL and DR must still use Standard regions** (VR-6)
6. Record exception in `OperationRecord` with approval metadata

Evidence Tag

[Undocumented – architectural judgement]

HC-10: CROSS_GEO_EXTENSION_PATH_APPROVED [NEW]

Category: Cross-Geo DR

Source: Production Readiness Review Section 27

Status: New governance control for cross-geography DR assignments

Definition

Any DR assignment to a region outside the customer's chosen geography must match an explicitly enumerated Cross-Geo Extension path in the active `PlacementPolicy`.

Rationale

Cross-geography DR introduces data residency, compliance, and latency considerations that require explicit approval. The engine does not autonomously select cross-geo DR regions; it only accepts pre-approved extension paths defined in the active `PlacementPolicy`.

Enforcement

DR assignments to Standard Capacity Regions in a **different geography** are rejected if no approved extension path exists for the source geography.

Approved Cross-Geo Extension Paths (Example)

Source Geography	Approved DR Geographies	Primary DR Region	Rationale
Middle East	Europe	[Amended v2.4] Weighted-selected Europe Standard region (Switzerland North = default example)	Middle East has only 2 regions (UAE North, Saudi Arabia Central) hosting Prod+CVAl co-located; DR is placed cross-geo in a weighted-selected Europe region (DR-020, PLC-010b). DEC-001 RESOLVED : ME DR is offered cross-geo into Europe.
Australia	Asia Pacific	Singapore	Regional pair fallback
India	Asia Pacific	Singapore	Regional pair fallback

Middle East Cross-Geo DR Placement (Active — DEC-001 RESOLVED, v2.4)

[Amended v2.4] This is now the active Middle East placement. DEC-001 is **RESOLVED**: Middle East DR is offered cross-geo into Europe. Prod and CVAl co-locate in a weighted-selected Middle East region (separate CRGs), and **DR is placed cross-geo in a weighted-selected Europe Standard region** (DR-020, PLC-010b, A-ME1). The Europe destination sizes DR max-not-sum (DR-017).

Inputs:

- Customer geography: Middle East
- Available Middle East Standard regions: UAE North, Saudi Arabia Central (2 only)
- **[Amended v2.4]** Middle East DR: offered cross-geo into a weighted-selected Europe Standard region (DR-020, PLC-010b); the generic `DR_NOT_OFFERED` flag (DR-014) does **not** apply to the Middle East

Active placement (v2.4 — cross-geo DR into Europe):

1. **Prod**: weighted `argmax(PS_Prod)` over {UAE North, Saudi Arabia Central} → e.g., UAE North
2. **CVAl**: co-located with **Prod** in the Middle East region (separate CRG; co-location ≠ capacity sharing, ENV-003) → UAE North
3. **DR**: **weighted selection over Europe Standard regions** → e.g., Switzerland North (default example) — cross-geo, using the weighted capacity model

Validation:

- The weighted-selected Europe DR region must pass **HC-1 through HC-10** including DR coverage floor (HC-6)
- If the selected Europe region fails any HC, fall back to the next-best weighted Europe region; if all fail, **reject with an ops alert** (VR-9)
- The engine does **not** silently select any Europe region outside the approved extension paths (HC-10, VR-11)

Validation Rules (VR)

- **VR-8 [Amended v2.4]:** For a Middle East geography, the seed **must** record a cross-geo `dr_region` in a **weighted-selected Europe Standard region** (DR-020, PLC-010b); the former `DR_NOT_OFFERED` outcome no longer applies.
- **VR-8a [Amended v2.4]:** The Middle East cross-geo DR region is chosen by the **weighted capacity model** over Europe Standard regions (not a fixed Switzerland North) via the approved Cross-Geo Extension path.
- **VR-11:** Cross-Geo Extension DR path must be explicitly approved in active `PlacementPolicy` **and** (for the Middle East) gated by a recorded DEC-001 approval.

Implementation

```
def resolve_dr_region(customer_geography, candidate_dr_region):
    policy = get_active_placement_policy()

    # DR-014: the generic DR_NOT_OFFERED flag still applies to any FUTURE geography
    # that carries it. [Amended v2.4] It no longer applies to the Middle East.
    if policy.dr_not_offered.get(customer_geography, False):
        return "NOT_OFFERED" # seed records no DR; no cross-geo substitution (VR-6)

    # [Amended v2.4] DR-020 / PLC-010b: the Middle East gets cross-geo DR in a
    # weighted-selected Europe Standard region (DEC-001 RESOLVED, A-ME1).
    if customer_geography == "Middle East":
        return weighted_select_dr_region(europe_standard_regions(), policy)

    return candidate_dr_region

def validate_HC10_cross_geo_extension(customer_geography, dr_region):
    policy = get_active_placement_policy()

    # DR_NOT_OFFERED short-circuits HC-10: no DR region to validate.
    if dr_region == "NOT_OFFERED":
        return PASS # legal no-DR outcome (DR-014, DEC-001)

    # Check if DR region is in same geography as customer
    if dr_region.geography == customer_geography:
        return PASS # Same-geo DR, HC-10 does not apply

    # Cross-geo DR → validate against approved extension paths
    approved_paths = policy.cross_geo_extension_paths.get(customer_geography, [])

    if dr_region not in approved_paths:
        return REJECT, f"HC-10 CROSS_GEO_EXTENSION_PATH_APPROVED: DR region
{dr_region.name} not in approved extension paths for {customer_geography}"

    # [Amended v2.4] DEC-001 RESOLVED: the Middle East cross-geo path is ACTIVE.
    # The DR region must be one of the approved weighted-selected Europe regions.
    return PASS
```

PlacementPolicy Configuration (Example)

```
{
  "policy_version": "2026-08-Q3",
  "dr_not_offered": {},
  "dr_not_offered_rationale": {
    "_comment": "[Amended v2.4] DEC-001 RESOLVED: Middle East DR is now offered cross-geo into Europe (DR-020). The generic DR_NOT_OFFERED flag remains available for any future geography but is not set for the Middle East."
  },
  "cross_geo_extension_paths": {
    "Middle East": {
      "approved_dr_regions": ["Switzerland North", "Sweden Central"],
      "selection_model": "weighted (DR-020, PLC-010b) over Europe Standard regions",
      "rationale": "[Amended v2.4] Active cross-geo DR path. Middle East Prod+CVAL co-locate in a weighted-selected Middle East region; DR is placed cross-geo in a weighted-selected Europe Standard region (Switzerland North is the default example, not fixed). A-ME1."
    },
    "Australia": {
      "approved_dr_regions": ["Singapore", "Japan East"],
      "rationale": "Regional pair fallback for Australia East / Australia Southeast"
    }
  }
}
```

Evidence Tag

[Undocumented – architectural judgement]

HC-11: AVAILABILITY_SET_INELIGIBLE [NEW — v2.4]

Category: Reservation Eligibility

Source: Requirements Baseline v2.4 (**CAP-020**); ADR-002 v2.4 (Capacity Reservation Model)

Status: Stable

Definition

A VM that is a member of an Availability Set is **INELIGIBLE** for a zonal on-demand capacity reservation and is rejected from the zonal reservation / per-AZ CRG path. $\text{Eligible}(\text{vm}) \Rightarrow \text{vm.availability_set} == \text{null}$ (i.e. the VM is zonal, not Availability-Set)

Rationale

An **Availability Set** and a **zonal capacity reservation** are mutually exclusive Azure placement constructs — an Availability-Set VM is not zone-pinned and cannot be backed by a zonal on-demand capacity reservation. Admitting such VMs to the zonal path would create reservations that can never be consumed. This is a **binary pass/fail eligibility gate** applied at VM onboarding, complementary to the region-level gates HC-1..HC-10.

Implementation

- Evaluated by the Onboarding Validator **before** zonal scoring/placement (TDD §8.1).
- **Reject** any VM where `availability_set` is set from the zonal reservation path; surface a structured onboarding error with remediation (migrate to a zonal deployment).

- Pairs with **CAP-021**: a non-zone-pinned but non-Availability-Set VM must first be **deallocated and redeployed into a target AZ** before it can associate with a per-AZ CRG reservation.
- **Applies to**: all environments (Prod, CVAL, DR) for the managed zonal reservation estate.

Examples

VM placement construct	Eligible for zonal reservation?
Zone-pinned VM (zone 1/2/3)	✓ Yes
Non-zonal VM, no Availability Set	⚠ Not until deallocate/redeploy-to-AZ (CAP-021)
Availability-Set member VM	✗ No (HC-11 / CAP-020)

Evidence Tag

[Decided] — Requirements Baseline v2.4 CAP-020.

Part 2 — Hard Constraint Summary Table

HC	Name	Type	Enforce- ment Stage	Primary Im- pact	Updated in Pass
HC-1	RE- GION_SEPAR ATION	Isolation	Stage 2	Prod isolated from Non- Prod/DR; NonProd=DR co-location allowed	Pass 2 (D8)
HC-2	CAPA- CITY_FLOOR	Capacity	Stage 2	Minimum CR headroom re- quired	Stable
HC-3	QUOTA_FLOOR	Quota	Stage 2	Minimum quota head- room per en- vironment type (Prod/ NonProd/DR)	Pass 2 (QG-3)
HC-4	DR_SEPARATI ON_CLASS	Resilience	Stage 2	DR region must have HIGH separa- tion class from Prod	Stable
HC-5	ZONE_AVAILA BILITY	Resilience	Stage 2	Minimum 2 availability zones re- quired	Stable
HC-6	DR_COVERAG E_FLOOR	DR Capacity	Stage 2	Combined DR+NonProd capacity must absorb DR demand	Pass 2 (D8)
HC-7	DR_FLOOR_IN TEGRITY	Quota Protec- tion	Stage 2	NonProd placement cannot encroach on DR quota floor	Pass 2 (QG-4)
HC-8	GEO- GRAPHY_CON TAINMENT	Geography	Stage 1	Prod region must be with- in customer's	Implied

HC	Name	Type	Enforce- ment Stage	Primary Im- pact	Updated in Pass
				chosen geo- graphy (Scenario 1)	
HC-9	STAND- ARD_REGION _ONLY	Region Class	Stage 1	Only Stand- ard Capacity Regions eli- gible for automated placement	PRR Section 27
HC-10	CROSS_GEO_ EXTEN- SION_PATH_A PPROVED	Cross-Geo DR	Stage 2	Cross-geo- graphy DR requires ex- plicit approval path	PRR Section 27
HC-11	AVAILABIL- ITY_SET_INEL IGIBLE	Reservation Eligibility	Onboarding (VM-level)	Availability- Set VMs ex- cluded from zonal reser- vations	Baseline v2.4 (CAP-020)

Part 3 — Enforcement Pipeline

Stage 1 — Pre-filtering (Eligibility Decision Tree)

Applied before candidate set formation:

```

All Azure Regions
↓
[HC-9: STANDARD_REGION_ONLY filter]
↓
Standard Capacity Regions
↓
[HC-8: GEOGRAPHY_CONTAINMENT filter] (if Scenario 1)
↓
Standard Regions in Customer Geography
↓
Candidate Set for Stage 2

```

Stage 2 — Hard Constraint Gate

Applied to each candidate region before scoring:

```
def apply_hard_constraints_stage2(region, environment_type, customer_context):
    """
    Returns: (PASS, None) or (REJECT, reason)
    """
    # HC-1: Region separation
    if not validate_HC1_region_separation(region, environment_type, customer_context):
        return REJECT, "HC-1 REGION_SEPARATION"

    # HC-2: Capacity floor
    if not validate_HC2_capacity_floor(region, environment_type):
        return REJECT, "HC-2 CAPACITY_FLOOR"

    # HC-3: Quota floor
    if not validate_HC3_quota_floor(region, environment_type, customer_context):
        return REJECT, "HC-3 QUOTA_FLOOR"

    # HC-4: DR separation class (DR only)
    if environment_type == "DR" and not validate_HC4_dr_separation_class(region, customer_context.prod_region):
        return REJECT, "HC-4 DR_SEPARATION_CLASS"

    # HC-5: Zone availability
    if not validate_HC5_zone_availability(region):
        return REJECT, "HC-5 ZONE_AVAILABILITY"

    # HC-6: DR coverage floor (DR only)
    if environment_type == "DR" and not validate_HC6_dr_coverage_floor(region, customer_context):
        return REJECT, "HC-6 DR_COVERAGE_FLOOR"

    # HC-7: DR floor integrity (NonProd only)
    if environment_type == "NonProd" and not validate_HC7_dr_floor_integrity(region, customer_context):
        return REJECT, "HC-7 DR_FLOOR_INTEGRITY"

    # HC-10: Cross-geo extension path (cross-geo DR only)
    if environment_type == "DR" and region.geography != customer_context.geography:
        if not validate_HC10_cross_geo_extension(customer_context.geography, region):
            return REJECT, "HC-10 CROSS_GEO_EXTENSION_PATH_APPROVED"

    return PASS, None
```

Stage 3 — Scoring Pipeline

Regions surviving HC-1..HC-10 enter scoring:

- **Prod:** Ranked by `PS_Prod(r)` = argmax over survivors
- **NonProd:** Ranked by `PS_NonProd(r)` = argmax over survivors (excluding Prod region)
- **DR:** Ranked by `PS_DR(r)` = argmax over survivors (excluding Prod region; may include NonProd region per HC-1)

Part 4 — Validation Rules (VR) Cross-Reference

The Production Readiness Review defines **11 Validation Rules (VR-1..VR-11)** that enforce hard constraints:

VR	Description	Enforced HCs
VR-1	Prod and DR must not share a region	HC-1
VR-2	CVAL and Prod must not share a region	HC-1
VR-3	CVAL and DR may share a region only under approved policy	HC-1 (D8 update)
VR-4	Derived Prod region must be within customer's chosen geography	HC-8
VR-5	All automated placement paths must use Standard Capacity Regions only	HC-9
VR-6	CVAL and DR must not use Restricted Capacity Regions under any condition	HC-9
VR-7	Standard region passes HC-1 through HC-10 → if all excluded, exhaustion error	HC-1..10
VR-8	[Amended v2.4] For Middle East geography, DR is placed cross-geo in a weighted-selected Europe Standard region (DR-020, PLC-010b, DEC-001 RESOLVED); the former DR_NOT_OFFERED outcome no longer applies	HC-10
VR-9	[Amended v2.4] If all weighted Europe Standard regions fail HC-1..HC-10 for a Middle East cross-geo DR placement, block placement with ops alert	HC-1..10
VR-10	Restricted region requested by customer → Scenario 2 exception path only	HC-9

VR	Description	Enforced HCs
VR-11	Cross-Geo Extension DR path must be explicitly approved in active PlacementPolicy	HC-10

Part 5 — POC Blockers & Testing

Critical POC Blockers

ID	Issue	Blocked HCs	POC Gate	Resolution
B-1	Azure Quota Groups functionality in target tenant/region	HC-3, HC-7	POC-30	GA availability check; if 404 → escalate to Azure Support
B-2	Quota pool release behavior on CR reduction (Tier 2/3 quota-neutral claim)	HC-7	POC-31	Measure release latency; confirm < 5 min for Tier RTO

POC Test Coverage

POC-01 through POC-10: Hard constraint validation (HC-1..HC-10)

POC-11: Middle East cross-geo DR (HC-10) — **[Amended v2.4] active** (DEC-001 RESOLVED); validates weighted DR selection over Europe Standard regions and max-not-sum sizing at the Europe destination (DR-020, PLC-010b)

POC-30: Quota Groups API integration (HC-3, HC-7)

POC-31: Quota pool release latency (HC-7 emergency transfer dependency)

Part 6 — Configuration & Policy

PlacementPolicy Schema (Hard Constraint Section)

```
{
  "policy_version": "2026-08-Q3",
  "hard_constraints": {
    "capacity_floor_multiplier": 2,
    "dr_bootstrap_qty": "configurable per workload/product/region (DR-007) – no fixed ratio",
    "dr_destination_sizing": "max-not-sum (A.6/DR-017) – MAX(source portions failing over to region)",
    "allow_single_zone_dev": false,
    "cross_geo_extension_paths": {
      "Middle East": {
        "approved_dr_regions": ["Switzerland North", "Sweden Central"],
        "selection_model": "weighted (DR-020, PLC-010b)",
        "rationale": "[Amended v2.4] Cross-geo DR into a weighted-selected Europe Standard region (DEC-001 RESOLVED)"
      }
    },
    "separation_class_overrides": {}
  }
}
```

Policy Constants

Constant	Default Value	Configurable?	Affected HCs
capacity_floor_multiplier	2	✓ Yes (per geography/SKU)	HC-2
dr_bootstrap_qty	(none — per workload/product/region)	✓ Yes (DR-007)	HC-6, HC-7
dr_destination_sizing	max-not-sum: MAX(source portions) (A.6/DR-017)	✓ Yes (SUM override per C-11)	HC-6, HC-7
~~dr_ratio_min / max / target~~	~~0.30 / 0.40 / 0.35~~	✗ Retired (Scenario 15 → 17)	—
min_zone_count	2	✓ Yes	HC-5
allow_single_zone_dev	false	✓ Yes	HC-5
cross_geo_extension_paths	{}	✓ Yes	HC-10

Part 7 — Implementation Checklist

Engineering Backlog Cross-Reference

HC	Primary Epic	Stories	Tasks
HC-1..HC-8	EPIC-07 (Regional Snapshot), EPIC-08 (Placement)	S0702, S0703, S0704, S0705	T070201
HC-9, HC-10	EPIC-07 (Regional Snapshot), EPIC-08 (Placement)	S0702	T070202
HC-3, HC-7 (Quota Groups)	EPIC-03 (Quota Management)	S0310, S0311, S0314	T031001..T031006
HC-6 (DR Coverage)	EPIC-06 (DR Management), EPIC-08 (Placement)	S0601, S0704	T060101, T070401

Pre-Deployment Validation

- [] HC-1..HC-10 validation functions implemented (unit tested)
- [] Stage 1 pre-filtering logic (HC-8, HC-9) validated
- [] Stage 2 hard constraint gate logic (HC-1..HC-7, HC-10) validated
- [] RegionalSnapshot schema extended with HC-6 fields (nonprod_crg_effective_free, dr_crg_free_slots)
- [] QuotaGroup integration (HC-3, HC-7) validated via POC-30
- [] **[Amended v2.4]** Cross-Geo Extension paths configured in PlacementPolicy (Middle East → weighted Europe Standard regions, DR-020/PLC-010b; DEC-001 RESOLVED)
- [] Middle East placement validated: Prod+CVAL co-located in a weighted-selected Middle East region; `dr_region` = weighted-selected Europe region (cross-geo)
- [] Middle East cross-geo DR validated: weighted Europe selection + max-not-sum sizing at Europe destination (HC-6/HC-7/HC-10, DR-017)
- [] Exception deployment workflow (HC-9 Restricted region override) tested
- [] HC rejection reasons logged to OperationRecord for audit
- [] Validation Rules VR-1..VR-11 traced to HC enforcement points

Appendix A — Decision Log Cross-Reference

Decision	HCS Affected	Change
D8	HC-1, HC-6	Removed <code>DR_region ≠ Non-Prod_region</code> constraint; added HC-6 to operationalise co-location
D1, D2	HC-3, HC-7	Introduced Two-Group Quota Architecture; HC-3 now reads from Quota Groups; HC-7 enforces DR floor
D9	HC-9, HC-10	Introduced Standard/Restricted region classification; added HC-9 and HC-10 governance gates

Appendix B — Evidence Tags

All hard constraints carry one of four evidence tags:

- **[Documented]** — Derived from Azure platform documentation (e.g., Quota Groups API)
- **[Derived]** — Logical consequence of a documented constraint or design decision
- **[Decided]** — Explicit design choice (traceable to Decision Log D1-D9)
- **[Undocumented — architectural judgement]** — Azure Well-Architected Framework best practice, not explicitly documented by Microsoft

Document Status: Production Ready

Next Review: After POC-30 and POC-31 completion (Quota Groups GA validation)